

CMPG 325 — COMPUTER NETWORKS

INDIVIDUAL SEMESTER PROJECT BRIEF

CMPG325-2026-023

Student	KOBE, LT
Student Number	45532206
Project ID	CMPG325-2026-023
Client ID	CLI-023
Assigned Organisation	Thuto Skills Training College (Klerksdorp)
Industry	Education

1. Your Role

You are responsible for analysing the assigned client scenario, designing and simulating the network in Cisco Packet Tracer, configuring your assigned networking challenge, testing the solution, and documenting evidence of your work.

2. Your Client

Client ID: CLI-023

Organisation: Thuto Skills Training College (Klerksdorp)

Industry: Education

3. Client Background

The approved project allocation identifies the organisation and industry above. Your solution must remain specific to this assigned client and must not be substituted with another project scenario.

4. Organisational / Network Context

Develop the network around the requirements, constraint, addressing block and change request specified in this brief. Use the main CMPG 325 Project Handbook for common project guidance.

5. Current Network Problem / Design Need

Design and simulate a computer network that addresses the client's networking needs. The completed network must permit successful data exchange between the appropriate nodes and must demonstrate the assigned networking challenge.

6. Client Requirements

- Assigned organisation: Thuto Skills Training College (Klerksdorp)
- Industry: Education
- Assigned addressing block: 10.18.0.0/16
- Provide appropriate connectivity and network services for the assigned scenario.
- Accommodate the stated design constraint and change request.
- Produce a working, testable Packet Tracer implementation.

7. Network Design Requirements

- Use Cisco Packet Tracer for the network implementation and simulation.
- Design an appropriate topology and device arrangement for the assigned scenario.
- Use the assigned addressing block as the basis of the IP addressing plan.
- Configure the necessary routers, switches, end devices and other required nodes.
- Demonstrate successful connectivity and testing.
- Document significant design decisions and evidence in GitHub.

8. Design Constraint

Boardroom requires dedicated wireless and wired presentation ports.

9. Assigned Networking Challenge

VLAN Trunking (802.1Q across switches)

You must configure, verify and demonstrate VLAN Trunking (802.1Q across switches) within your assigned client network. Be prepared to explain what you configured, why it is appropriate, and how you verified that it works.

Difficulty classification: Advanced CMPG 325

10. Client Change Request

CR9: One off-site administrator requires secure remote management access to network devices.

Your final solution must show how the network accommodates this change request. Follow the specific wording of this project rather than introducing additional scope.

11. Testing Requirements

- Test relevant end-to-end connectivity.
- Verify the assigned networking challenge.
- Capture clear evidence of successful operation/configuration.
- Document important troubleshooting performed.
- Ensure the final .pkt file opens and reproduces the working solution.

12. GitHub Portfolio of Evidence

- Create and maintain an individual GitHub portfolio.
- Include a professional README and appropriate project evidence.
- Organise evidence for client requirements, network design, IP addressing, topology, Packet Tracer, configuration, testing, screenshots, troubleshooting and reflection as appropriate.
- Use meaningful commits that show development over time.
- Do not submit another student's work.

13. Individual Video Demonstration

- Submit a 15–20 minute individual video with an inset webcam view.
- Introduce yourself and the assigned client/project.
- Explain client requirements and your network design.
- Demonstrate the Packet Tracer implementation.
- Demonstrate your assigned networking challenge.
- Show relevant configuration and connectivity/testing.
- Discuss important design decisions and troubleshooting.
- Provide a short reflection on the completed project.

14. Project Milestones

- Project commencement: 14 August 2026
- Milestone 1: 28 August 2026
- Milestone 2: 2 October 2026
- Final submission: 16 October 2026

Use the official CMPG 325 project workflow and lecturer-issued milestone guidance. Do not invent alternative deadlines.

15. Final Submission Requirements

- Packet Tracer .pkt file
- GitHub portfolio/link
- Required technical documentation and evidence
- 15–20 minute individual video demonstration
- Final submission through eFundi

16. Project-Specific Notes

Project ID: CMPG325-2026-023

Client ID: CLI-023

Organisation: Thuto Skills Training College (Klerksdorp)

Technical challenge: VLAN Trunking (802.1Q across switches)

Addressing block: 10.18.0.0/16

Constraint: Boardroom requires dedicated wireless and wired presentation ports.

Change request: CR9: One off-site administrator requires secure remote management access to network devices.

Academic Integrity & NWU AI Policy

You remain responsible for the correctness, understanding, verification and academic integrity of everything you submit. Any AI use must comply with the applicable NWU AI Policy. AI assistance does not transfer responsibility for the submitted work from the student.

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